

10 (**short**) A reference volume V_0 of ceramic subjected to a uniform tensile stress σ has a survival probability

$$P_s(V_0) = \exp \left[- \left(\frac{\sigma}{\sigma_0} \right)^m \right]$$

where σ_0 and m are constants.

(a) Explain briefly why the values of σ_0 and m depend on the specimen volume V_0 . [2]

(b) A batch of identical ceramic rods are tested to failure in tension. The rods are solid circular cylinders, loaded parallel to the axis. 85% of the specimens survive a tensile stress $\sigma = 300$ MPa, and 20% survive $\sigma = 400$ MPa. Calculate the constants σ_0 and m for the rods. [4]

(c) A number of the rods described in (b) are joined together end-to-end using an adhesive, applied uniformly across the circular faces, to create a longer rod of the same diameter. This is loaded in tension parallel to the cylinder axis. If the adhesive fails at a tensile stress $\sigma = 290$ MPa, how many rods can be joined together before failure is more likely to occur in the ceramic than the adhesive? [4]